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Policy Measures to Enhance Manufacturing Data Utilization for AI Transformation (AX) in Manufacturing Firms: A Supply Chain-Based Approach

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SUMMARY

Policy Measures to Enhance Manufacturing Data Utilization for AI Transformation (AX) in Manufacturing Firms: A Supply Chain-Based Approach

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■ Need to Enhance Manufacturing Data Utilization for Manufacturing AX

- The Role of Manufacturing Data as the Foundation of Manufacturing AX

  - Manufacturing data refers to data generated, collected, and utilized throughout the entire manufacturing lifecycle, including product planning, design, production, quality management, distribution, sales, and maintenance.

  - In the era of manufacturing AX, manufacturing data has evolved from simple production records into a strategic asset that enables defect prediction, predictive maintenance, process optimization, production planning, energy efficiency, and supply chain risk management.

  - Although manufacturing SMEs are the primary generators and holders of manufacturing data, their AI adoption remains constrained by insufficient data infrastructure, limited analytical capabilities, and weak linkages between production processes and data.

- Characteristics of Manufacturing Data from an AI Perspective

  - Manufacturing data is domain-specific data generated and interpreted within the context of production processes, equipment, materials, and shop-floor environments, requiring high levels of accuracy, reliability, traceability, temporal synchronization, and secure access control for effective AI applications.

- Limited availability of rare-event data and changes in data distributions caused by evolving production processes hinder the deployment, continuous 34 Summary updating, and retraining of AI models.

- Enhancing Manufacturing Data Utilization from a Supply Chain Perspective

  - Growing requirements for product-level data traceability driven by global supply chain restructuring, trade and environmental regulations, and initiatives such as the Digital Product Passport (DPP) and the Carbon Border Adjustment Mechanism (CBAM) have increased the importance of inter-firm data connectivity.

  - Enhancing manufacturing data utilization requires policy approaches that go beyond supporting individual firms and instead consider data flows and collaborative relationships among equipment suppliers, solution providers, neighboring manufacturing firms, and large anchor companies within supply chains.

Manufacturing Data Utilization and Sharing Cases: Policy Implications

- Data Platform Cases and the Foundation for Manufacturing Data Utilization

  - Data platforms have improved accessibility to manufacturing data while supporting industrial data integration and exchange through datasets, analytical tools, cloud services, and data marketplaces.

  - However, simply providing datasets, analytical tools, and cloud infrastructure is insufficient to ensure practical implementation, continuous AI model retraining, and feedback into manufacturing processes.

  - To expand platform-based support toward supply chain-wide data collaboration, institutional arrangements for data quality, standards, metadata, usage conditions, transaction rules, and benefit-sharing mechanisms are required.

- Data Space Cases and Supply Chain Data Collaboration

  - Data space initiatives demonstrate the importance of data sovereignty, access control, interoperability, standardization, and trust-based governance for supply chain collaboration.

  - Establishing data spaces alone does not guarantee effective data sharing or utilization, as differences in internal systems, production data structures, and data quality among participating firms continue to limit practical implementation.

  - Supply chain collaboration therefore requires trusted intermediary mechanisms, including data linkage standards across equipment, processes, and solutions, privacy-preserving data-sharing methods, data trusts, and standardized contractual arrangements.

■ Supply Chain-Based Manufacturing Data Ecosystem and Collaboration Types

- Classification of Supply Chain Collaboration Based on Data Flows

  - Collaboration types are classified not by firm size or industrial sector but by the relationships through which manufacturing data is generated, owned, analyzed, and shared around manufacturing SMEs.

  - Equipment suppliers generate production and equipment data; solution providers collect and analyze manufacturing data for AI applications; neighboring manufacturers collaborate around common production challenges; and large enterprises connect production planning, quality, delivery, and demand information across supply chains.

- Bottlenecks and Improvement Directions by Collaboration Type

  - (Manufacturing SMEs-Equipment Suppliers) Limited access to equipment data, heterogeneous data formats, and insufficient data acquisition capabilities in legacy equipment highlight the need for standardized and accessible equipment data.

- (Manufacturing SMEs-Solution Providers) Discontinuous utilization after system deployment, insufficient interoperability between existing systems and new AI solutions, and weak AI model retraining require stronger operational support and continuous feedback mechanisms.

- (Neighboring Manufacturing SMEs) Limited rare-event data, insufficient trust among firms, and unclear benefit-sharing and liability arrangements call for collaborative data-sharing frameworks focused on joint problem solving.

- (Manufacturing SMEs-Large Enterprises) One-way data collection and limited feedback of analytical insights require bidirectional supply chain data sharing and fair governance mechanisms for manufacturing data utilization.

■ Policy Measures to Enhance Supply Chain-Based Manufacturing Data Utilization

1. (Manufacturing SMEs-Equipment Suppliers) Institutional and Technical Support for Equipment Data Utilization

- Support demonstration projects for domestically developed manufacturing equipment incorporating manufacturing data standards

  - Verify not only equipment performance but also manufacturing data acquisition capabilities, standardized data elements, and interoperability with external systems during demonstration projects.

  - Promote joint demonstration projects involving equipment suppliers and manufacturing SMEs to validate linkages between equipment, process, and quality data for AI applications.

- Establish support systems for equipment data quality verification and data migration

  - Develop equipment data quality reporting systems covering accuracy, missing values, collection frequency, and synchronization with production data.

- Provide technical support for migrating and integrating historical operational, maintenance, and process data when replacing or upgrading manufacturing equipment.

- Protect data creators' rights and establish incentives for data utilization

  - Clarify ownership and usage rights for manufacturing data generated through equipment operation.

  - Link manufacturing data valuation with incentive mechanisms supporting data provision, quality improvement, and data-driven maintenance activities.

2. (Manufacturing SMEs-Solution Providers) Continuous Operation and Integration of Manufacturing AI Solutions

- Standardize data structures and interoperability across manufacturing solutions

  - Develop guidelines for standardized data structures across MES, ERP, quality management, energy management, and AI solutions.

  - Include interoperability with existing systems as a key criterion in government-supported AI solution programs.

- Strengthen solution monitoring and data preprocessing support

  - Establish monitoring systems for data collection status, AI model performance, and utilization outcomes.

  - Expand technical support for data cleansing, labeling, quality assessment, and preprocessing.

- Develop in-house operational capabilities and link data valuation with financial support

  - Support workforce development for continuous operation of manufacturing AI solutions.

- Connect manufacturing data valuation with financing and guarantee programs.

3. (Neighboring Manufacturing SMEs) Collaborative Data-Sharing Mechanisms

- Establish common data taxonomies and minimum exchange standards

  - Develop common classifications for process, defect, and equipment data.

  - Define minimum data elements required for collaborative analysis while protecting trade secrets.

- Support privacy-preserving collaborative AI models

  - Promote collaborative AI models that enable joint learning without sharing raw data.

  - Support demonstration platforms addressing common manufacturing challenges.

- Develop data trust mechanisms and benefit-sharing frameworks

  - Establish data trusts responsible for data management, access control, and legal accountability.

  - Introduce benefit-sharing mechanisms reflecting firms' contributions to data provision and AI performance improvement.

4. (Manufacturing SMEs-Large Enterprises) Bidirectional Supply Chain Data Sharing and Fair Data Governance

- Strengthen international data standards for global supply chain compliance

  - Improve interoperability with international standards for quality, delivery, demand, sustainability, and traceability data.

  - Clarify standardized data formats and verification requirements for SMEs.

- Support platforms for feeding analytical insights back to suppliers

  - Establish bidirectional data platforms through which large enterprises return analytical insights to SMEs.

  - Support SMEs in utilizing these insights for production planning and quality management.

- Develop standard contracts and fair governance for manufacturing data

  - Establish standard contractual guidelines covering data ownership, usage rights, analytical outputs, and third-party access.

  - Prevent unfair business practices associated with manufacturing data provision.

5. (Common and Long-Term Policy Agenda) Building a Sustainable Manufacturing Data Ecosystem

- Standardization and interoperability across supply chains

  - Expand manufacturing data standards to support end-to-end supply chain integration.

  - Develop mapping frameworks between domestic and international manufacturing data standards.

- Strengthen collaborative platforms and AI workforce development

  - Build trusted data collaboration platforms involving equipment suppliers, solution providers, SMEs, and anchor firms.

  - Foster AI professionals with integrated IT-OT capabilities while supporting in-house training, consulting, and learning communities.

- Establish institutional foundations for data rights, valuation, and benefit sharing

  - Clarify data ownership and usage rights and establish dispute resolution mechanisms.

  - Build frameworks for data valuation and fair benefit-sharing.